Trading on Attention: Can Wikipedia Pageviews Predict Daily Stock Returns? A Walk-Forward Test on 2016–2019 Data

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Independent portfolio project, extending a case from the AI for Business & Finance Certificate
(Columbia Business School Executive Education × Wall Street Prep)

Status: model and analysis plan fixed before running the test period; results to be added.

Problem Statement

Wikipedia records how many times each article is read every day. For a listed company, these pageviews are a free, daily measure of how much attention the company receives. The question behind this project is whether that attention contains information about the company's stock return in the following days.

The project builds on the "Trading with AI" case of the AI for Business & Finance Certificate. The case uses daily Wikipedia pageviews for 50 large S&P 500 stocks to predict next-day returns, builds a portfolio from the predictions, and compares it with the S&P 500 (SPY) over 2017. A single test year is a narrow basis for a conclusion: 2017 was a year of strong gains and low volatility, and the comparison uses SPY as the only benchmark.

This project reimplements the case with independently downloaded data, keeps the model unchanged, and extends the study to January 2016–December 2019, with a three-year test period and a stricter evaluation. We therefore ask:

> *Research question: Do lagged Wikipedia pageviews improve out-of-sample predictions of next-day returns for large U.S. stocks beyond lagged returns alone, and does a portfolio built on these predictions hold up against fair benchmarks, transaction costs, and random stock selection?*

Model Description and Methods

Overview

The model is a stock-ranking strategy. For each stock, a separate linear regression predicts the day's return from the previous week. Every trading day the 50 predictions are ranked and the portfolio holds the ten stocks with the highest predicted return. Three nested models add attention information step by step, so the contribution of pageviews can be isolated (Figure 1). The models are re-estimated every month using only past data.

Model Specification

Universe and data. The universe is 50 large S&P 500 stocks and, for each, four competitors, as defined in the original case, where the list was generated with ChatGPT. Daily prices for the 50 stocks and SPY are downloaded with `yfinance`, and daily pageviews of each company's English Wikipedia article are taken from the Wikimedia Pageviews API; article redirects (e.g., "The Home Depot" → "Home Depot") are resolved through the Wikipedia API. Data after 2019 are excluded to keep the COVID-19 pandemic out of the sample.

Variables. The daily return of stock i is

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}.$$

Attention enters in two forms: raw pageviews $v_{i,t}$, and proportional views, the stock's share of the views of its competitor group C_i (the stock and its four competitors),

$$s_{i,t} = \frac{v_{i,t}}{\sum_{j \in C_i} v_{j,t}}.$$

Every predictor is a seven-day average lagged by one day, so day t uses only information up to $t-1$:

$$\bar{x}_{i,t} = \tfrac{1}{7} \textstyle\sum_{k=1}^{7} x_{i,t-k}, \quad x \in \{r, v, s\}.$$

Prediction models. For each stock, three ordinary least squares models are estimated:

$$\begin{aligned}
&\text{M1: } r_{i,t} = \alpha + \beta_1 \bar{r}_{i,t} + \varepsilon_{i,t} \\
&\text{M2: } r_{i,t} = \alpha + \beta_1 \bar{r}_{i,t} + \beta_2 \bar{v}_{i,t} + \varepsilon_{i,t} \\
&\text{M3: } r_{i,t} = \alpha + \beta_1 \bar{r}_{i,t} + \beta_2 \bar{v}_{i,t} + \beta_3 \bar{s}_{i,t} + \varepsilon_{i,t}
\end{aligned}$$

M1 uses past returns only; M2 and M3 test whether attention adds information to it.

Trading rule. On each test day t, model m produces a predicted return $\hat{r}^m_{i,t}$ for all 50 stocks. The portfolio holds the top ten in equal weights for one day, giving the strategy return

$$R^m_t = \tfrac{1}{10} \textstyle\sum_{i \in \mathrm{Top}_{10}(\hat{r}^m_{\cdot,t})} r_{i,t}.$$

The strategy is long-only and rebalanced every day.

Walk-forward design. The sample runs from January 2016 to December 2019. Pageview data are available from July 2015, but the sample starts in 2016 because the training window is 12 months: a full calendar year (January–December 2016) forms the first training window, as in the original case, and the second half of 2015 would not fit a

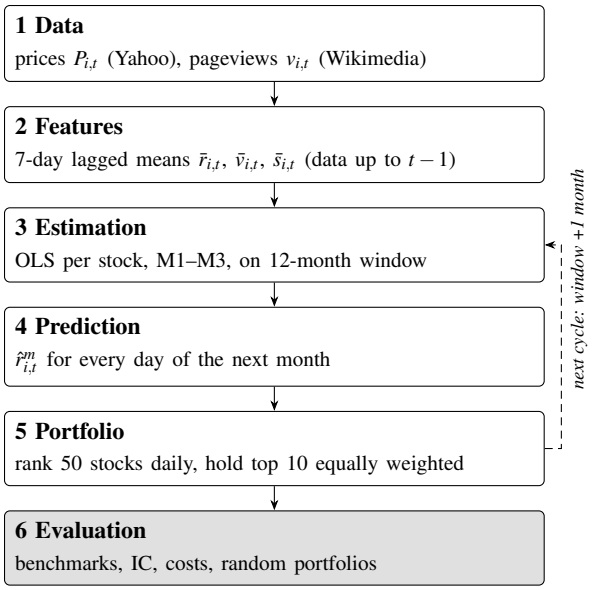


Figure 1: Pipeline of one walk-forward cycle. Steps 3–5 repeat for each of the 36 monthly cycles; step 6 is applied once to the full 2017–2019 test period.

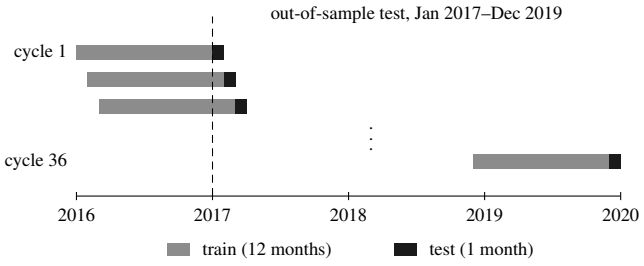


Figure 2: Walk-forward design. Each cycle trains on 12 months and tests on the next month; the window rolls forward one month at a time.

complete window. Only the last days of December 2015 are used, to compute the lagged seven-day averages for the first week of January 2016. Estimation follows a rolling walk-forward scheme (Figure 2). Each cycle fits the models on a 12-month window and predicts the following month; the window then moves forward by one month. The first window covers January–December 2016 and predicts January 2017; the last covers December 2018–November 2019 and predicts December 2019, giving 36 monthly cycles and a three-year out-of-sample test. The first 12 cycles are the 2017 test of the original case and therefore validate the reimplementation.

Measures.

(1) cumulative and annual return, volatility, Sharpe ratio, maximum drawdown, and hit rate of each strategy;

(2) comparison with two benchmarks: SPY and an equal-weight portfolio of the 50 stocks;

(3) information coefficient (IC): the daily Spearman correlation between predicted and realised returns across the 50 stocks, and its average over the test period;

(4) stability of β_2 and β_3 across the 36 cycles, with confidence intervals;

(5) daily turnover and cumulative return at 0, 5, 10, and 20 bps of transaction costs;

(6) position of each strategy's return within the distribution of thousands of random ten-stock portfolios from the same universe;

(7) results reported separately for 2017, 2018, and 2019.

Motivation of Design Choices

The model is kept identical to the original case on purpose. The aim is to test whether the case's result holds up, not to find a better signal. Changing features, window lengths, or portfolio size after looking at the test period would make the results no longer out-of-sample. All extensions are therefore in the evaluation, not in the model.

The data are split in time order rather than at random. As the case itself shows, a random split mixes later days into the training set to predict earlier ones, which means using the future. Monthly re-estimation keeps the coefficients current, instead of trading in late 2019 with a model estimated on 2016 data. The 2017–2019 window covers three different years: strong, calm gains in 2017, a sharp fall in the fourth quarter of 2018, and a recovery in 2019, while leaving out the pandemic.

SPY alone is not a fair benchmark. Choosing ten stocks from a fixed list of 50 large companies mixes the effect of the list with the effect of the signal; an equal-weight portfolio of the same 50 stocks separates the two. The IC measures predictive power directly, independent of how the portfolio is built. Rebalancing ten positions every day has a cost, so results are shown at several cost levels. Random portfolios show whether the strategy's return could have come from chance.

Analysis Plan

The analysis proceeds in three steps. First, the reimplementation is checked by comparing the 2017 results of the first 12 cycles with the original case. Second, the full 36-cycle test is run with all choices in Table 1 fixed beforehand. Third, measures (1)–(7) are computed and presented in a cumulative return chart against both benchmarks, a performance table, a chart of returns by year, an IC chart, a chart of coefficients across cycles, a transaction-cost chart, and a histogram of random portfolios.

Table 1: Model parameters, values, and justification.

Parameter	Symbol	Value	Justification
Universe	N	50 stocks	Original case
Competitor group	C_i	stock + 4 competitors	Original case; defines proportional views $s_{i,t}$
Data period	–	Jan 2016–Dec 2019	Starts with the first full 12-month training window; 2020 onward excluded (COVID-19)
Feature window	–	7 days, lagged 1 day	Original case; uses data up to $t-1$ only
Training window	–	12 months, rolling	Original case; keeps coefficients current
Test window	–	1 month	Original case; monthly re-estimation
First training window	–	Jan–Dec 2016	12 months, as in the original case
Test period	–	Jan 2017–Dec 2019	Three years out-of-sample
Cycles	–	36	Out-of-sample Jan 2017–Dec 2019; first 12 reproduce the case
Estimator	–	OLS per stock	Original case; nested models M1–M3
Portfolio size	K	10, equal weight	Original case
Holding period	–	1 day	Original case; daily rebalancing
Benchmarks	–	SPY; equal-weight 50	Market, and the same 50-stock universe
Transaction costs	c	0, 5, 10, 20 bps	Sensitivity of results to costs
Random portfolios	–	thousands	Chance benchmark for picking 10 of 50